Solution Tutorial 8 (core exercises)

1. **General least-squares estimates**
   In this exercise we will use the Levenberg-Marquardt method in order to fit several different models.
   (a) Let's consider that we have a model that follows the following curve:

$$f(x) = \frac{a}{x} + bx + c \tag{1}$$

   (b) Generate 1,000 realizations of this curve with $(a = 2, b = 1, c = 2)$, for 20 points evenly spaced between 0.5 and 4. On top of the curve add white Gaussian noise with $\sigma = 2$ and use the Levenberg-Marquardt method to find the a, b, c parameters.
   (c) Plot the realizations, the model and the best fit you obtained.
   (d) Repeat the exercise with a Gaussian noise of $\sigma = 1$, but instead of only using 20 evenly spaced points use 10, 20 and 30 evenly spaced points and compare how well we still can retrieve the model.

Solution:

(a) In Figure 1 we have plotted the function $f(x)$ (dashed line).

(b) For the least-square estimation the problem is reduced to solve $(A^T A)p = A^T y$ for p (slide 13 in Lecture 8). In this case the matrix A is given by:

$$A = \begin{bmatrix} 1/x_1 & x_1 & 1 \\ 1/x_2 & x_2 & 1 \\ ... & ... & ... \\ 1/x_{20} & x_{20} & 1 \end{bmatrix}$$

with $x_1 ... x_{20}$ the 20 evenly spaced x points, and the vector $p = [a, b, c]$. Here $y = [y_1, y_2, ..., y_{20}]$ with y_i the value of the function ($f(x)$ evaluated at x_i and with $a = 2$, $b = 1$, $c = 2$) plus the Gaussian noise (to add Gaussian noise for each x point, you need to randomly take a number from a Gaussian with the specified sigma value, and add it to the f(x) evaluated at that point). Solving this system of equations give us the value of the parameters a, b, c for each realization.

(c) In Figure 1, we have plotted one realization (data points) and the best fitted curve obtained via least-square estimation. We also plotted the original model $f(x)$. In Figure 2, we show the best fit for 100 realizations.

(d) In Figure 3, we show one realization and the best fit for each of the three sigma values. For the case with $\sigma = 0.5$ (blue) we recover the original model (black dashed line) very accurately, but as long as the included noise in the data is larger, the best fitted curve is less accurate. We repeat this 10,000 times, and obtained the parameters in each case. The distribution of the three parameters for the three different noises are shown in Figure 4. As expected, the fitting for the measurements with the smaller noise $\sigma = 0.5$ (blue) results in a better recovery of the "perfect" model and the fitted parameters are most of the time very close to the real values for the three parameters. On the other hand, the fitting for the measurements with the larger noise $\sigma = 2.0$ (red) results in very inaccurate parameters for several of the realizations.

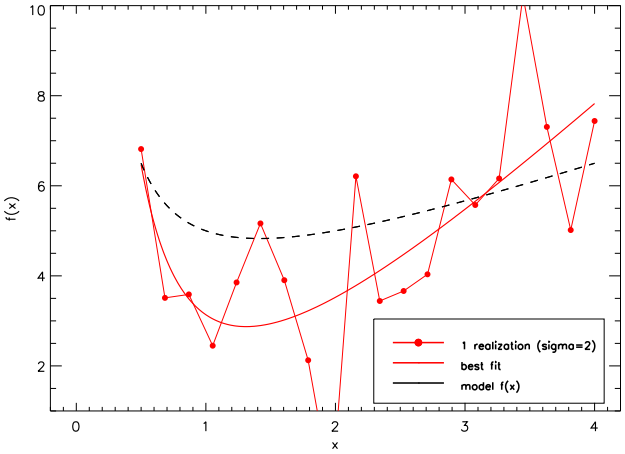

Figure 1: Red points are one realization of the curve, including Gaussian noise with $\sigma = 2$. The red solid line show the best fitting and the dashed black curve is the model with the parameters $a = 2$, $b = 1$, $c = 2$.

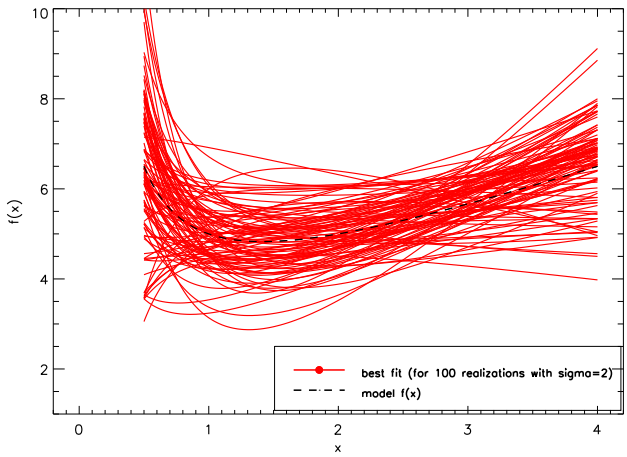

Figure 2: The best fit for 100 realizations of the model with $\sigma = 2$.

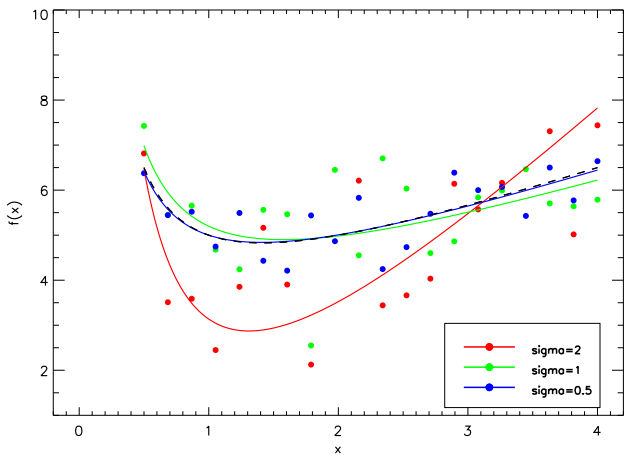

Figure 3: Red, green and blue points are one realization of the curve, including Gaussian noise with $\sigma = 2, 1.0$ and 0.5 respectively. The solid lines show the best fitting in each case and the dashed black curve is the model with the parameters $a = 2$, $b = 1$, $c = 2$.

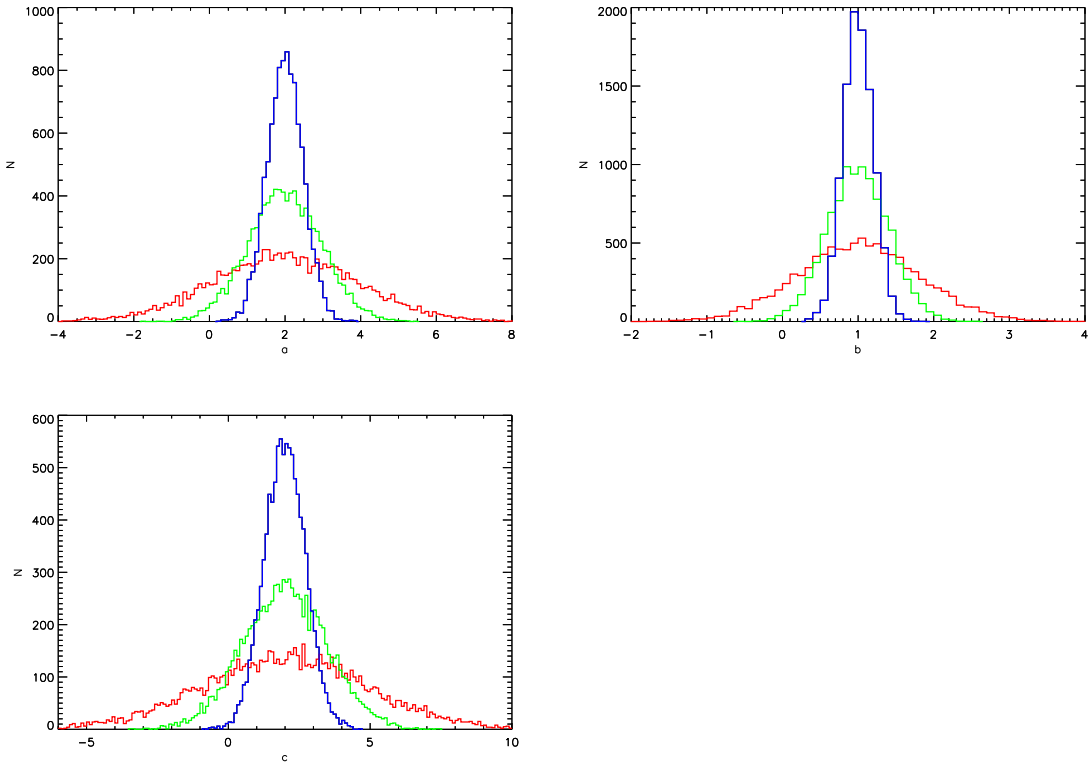

Figure 4: Distribution of the three parameters after to perform 10,000 realization of the data with the three different noises. Red, green and blue correspond to simulated data with Gaussian noise $\sigma = 2, 1.0$ and 0.5 respectively.

2. **Galaxy stellar mass function**
The galaxy stellar mass function describes how many galaxies there are with a certain stellar mass, per unit volume. In this exercise we will use a data set with the observed stellar mass values of galaxies for 5 galaxy clusters at redshift 0.101. The dataset can be downloaded from `https://home.strw.leidenuniv.nl/~daalen/NURfiles/smfdata.txt`, and it contains the stellar mass (m_*) in $\log_{10}$ units.

   (a) Construct the stellar mass function (SMF) for this data, which is the number of galaxies per mass bin per cluster. Use a convenient number of bins for $\log_{10} m_*$ range [8.5, 12].

   (b) The stellar mass function can be represented by the Schechter function as:

$$\Phi\left(\text{cluster}^{-1}\text{dex}^{-1}\right) = \ln(10) \cdot \Phi^* \cdot \left(10^{(M-M^*)(1+\alpha)}\right) \exp\left(-10^{(M-M^*)}\right) \tag{2}$$

   where $M \equiv \log_{10}(m_*)$. The parameters $M^* \equiv \log_{10}(m_*^*)$ is the characteristic mass, α is the low-mass slope, and Φ^* is the overall normalisation. Fit the data to obtain these parameters. Think about how to do the fitting! Plot your fitted line along with the SMF and your fitted parameters.

Solution:

(a) The stellar mass function (SMF) for this data is shown in the figure below. Number of bins was chosen via Freedman–Diaconis rule, and equals 20 in this case. Given that this is a counting process (counting galaxies per mass bin), we have included Poisson error bars to the data points.

(b) As this represents a counting process, a Poisson likelihood is the most adequate way to fit the data. First, compute the natural logarithm of the Poisson likelihood (slide 20 in Lecture 8). To do this, use the data (the data points in the figure) and compute the likelihood over a grid of different α, M^*, and ϕ^* values. You can then maximise the likelihood to obtain the best-fitting parameters. In this case, you cannot use Levenberg-Marquardt because it works only for χ^2. But you have several options: use the minimisation routine from the previous tutorial, or update your Levenberg-Marquardt for the case of log likelihood of Poisson distribution (replace χ^2 to $-\ln \mathcal{L}(\mathbf{p})$, as well as corresponding α and β).

We obtain the best fitted parameters to be $\alpha = -1.35$, $M^* = 11.36$, and $\phi^* = 10.28$, and we plot it as a solid black line. Note that when you change the number of bins, best-fit parameters can change significantly (try it yourself with 30 and 15 bins). For comparison, we also performed the fitting using a Gaussian likelihood, which results in $\alpha = -1.36$, $M^* = 11.32$, and $\phi^* = 10.32$, plotted as a dashed line. You can see that this results in slightly different parameters, and from the figure, you can see that this is less accurate in this case.

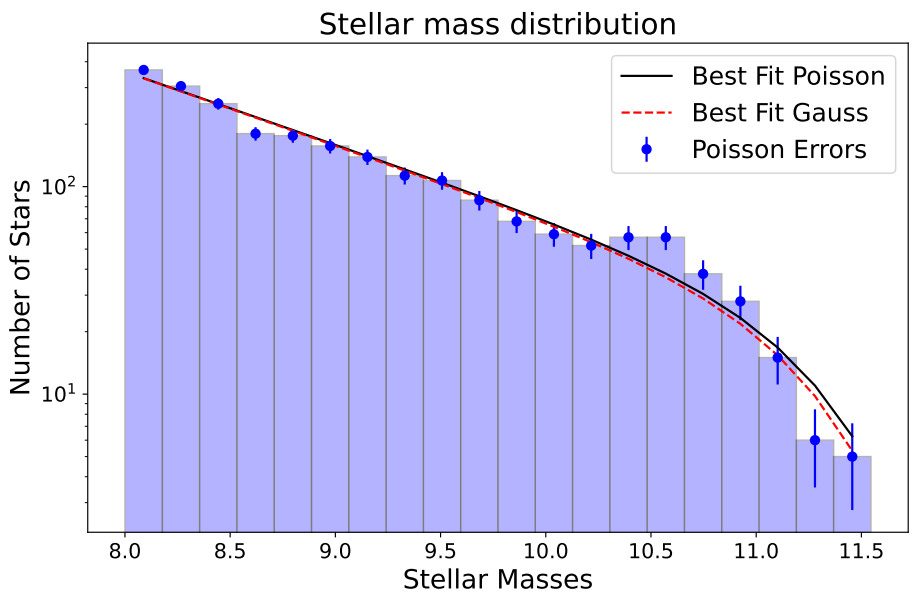

Figure 5: The SMF with the best Schechter fitted curve obtained maximising a Poisson likelihood (solid line) and a Gaussian likelihood (dashed line).